

Yongjun Rong

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EDUCATION

Jun.,2006, Master of Computer Science, Texas Tech University, Lubbock, TX, USA

Feb.,1996, Master of Engineering, Shanghai JiaoTong University, Shanghai, China

Jul.,1991, Bachelor of Engineering, Changsha Railway University, Changsha, Hunan, China

SUMMARY:

Yongjun Rong is a hands-on visionary cloud architect and technology leader with deep expertise in GenAI systems, prompt engineering, cloud infrastructure, and digital transformation. With over two decades of experience spanning finance, consulting, SaaS, and technology sectors, he has architected and delivered large-scale distributed systems, led mainframe-to-cloud migrations, and pioneered automation and DevOps best practices. Yongjun excels at designing resilient, scalable microservices, integrating enterprise SaaS applications, and driving innovation through mentorship and cross-team enablement. His technical breadth covers AWS, GCP, Azure, Java, Python, Node.js, Kubernetes, Terraform, and more. Fluent in both English and Chinese, he is passionate about empowering organizations to achieve agility, reliability, scalability and business impact through modern cloud-native solutions.

SKILLS:

- GenAI system architecture and implementation with Retrieval-Augmented Generation (RAG) and multi-agent orchestration to accelerate software development and mainframe migration to modern cloud-native platforms.
- Cloud infrastructure design, automation, and migration across AWS, GCP, and Azure, including serverless architectures (Lambda, EventBridge, Step Functions), container orchestration (EKS, AKS, GKS), and Infrastructure as Code (Terraform, Terragrunt, CloudFormation, Ansible, Puppet, Chef).
- Enterprise architecture leadership for large-scale finance, insurance, and telecommunications organizations, specializing in distributed systems, microservices, event-driven architectures, streaming data processing, and technology strategy development.
- Digital transformation engineering, refactoring legacy monolithic systems to cloud-native, loosely-coupled distributed architectures using Java, Python, Node.js, SpringBoot, and modern data stores (SQL, NoSQL, DynamoDB, MongoDB, ClickHouse, PostgreSQL, Oracle).
- DevOps and automation expertise in cloud and on-premises environments, including CI/CD pipeline design, configuration management, system/network administration, and orchestration with Docker, Kubernetes, Jenkins, Bamboo, and Linux HA technologies.
- Integration architecture and implementation for SaaS enterprise applications (ERP, CRM, Accounting, HR), leveraging VPC Endpoints, PrivateLink, REST/SOAP APIs, WSDL/XML/JSON, and secure authentication/authorization mechanisms.
- Design and delivery of highly available, scalable, and resilient microservices and event-driven systems, applying best practices in CQRS, event sourcing, domain-driven design, and cloud-native patterns for internet-scale applications.
- Mentorship and leadership in technology adoption, process automation, and cross-team enablement, fostering a culture of innovation, reliability, and continuous improvement across engineering organizations.

HOBBIES

- Pickleball (~4.0 player)
- Pingpong (ITTF Score around 1200)
- IoT or Robot programming (Have iRobot and MIPOSaur at home. Programming against iRobot create SDK and MIPOSaur SDK.)
- iOS/Android Apps Development (Developing a Chinese character learning App)
- Playing with cutting-edge technologies (genAI/Autogen/Multiagents, Docker, Go, Kafka, Kubernetes and Amazon CloudFormation, JavaScript, AngularJS, ReactiveJS, Node.js)
- IFTTT Home Automation (IFTTT, WeMo, EcoBee, Weather Underground, SaaS, REST, Home Automation)

WORK EXPERIENCE SUMMARY:

Employments	Major Accomplishments & Technologies
Feb.,2025 – Current Biophy Cherry Hill, NJ (WFH),USA Director of Cloud Architecture	- Comprehensive Infrastructure as Code (IaC) Design and Implementation - AI-Powered Data Ingestion and Scraping System for RAG - Custom AWS CloudWatch Alerting System with EventBridge, Lambda, SQS and SNS IaC, Terraform/Terragrunt, AWS/EC2/EBS/S3/VPC/ALB/Security/R53/IAM/Secrets Manager/Lambda/CloudWatch/EventBridge/SQS/SNS, Python/Django/Pydantic, openAI API/LangChain, SOLID Principles, Clean Architecture,
Dec.,2021 – Feb.,2025 McKinsey & Company Cherry Hill, NJ (WFH),USA Senior Cloud Engineer II (Associate)	- Using genAI (openAI) with RAG (ChromaDB, Milvus) to accelerate the development of data pipelines running on pyspark and databricks - Using genAI (openAI) with multi-agents to accelerate the migration of mainframe to modern technologies - Migration/Refactoring of Mainframe (IBM MQ) EDA system to AWS by using EventBridge/Lambda/Step Function/CloudWatch/EKS/SQS/SNS...etc. genAI/openAI/Langchain/RAG/Multi-agents/prompts tuning, VectorStore/ChromaDB/Milvus, Python/Flask/FastAPI/Pydantic/Jupyter, Databricks/pyspark, Typescript/JavaScript/NodeJs, Event Driven Architecture (EDA), Terraform, Microservices, Java/SpringBoot/NodeJs/Python/Flask/FastAPI, EventBridge/CloudWatch/Lambda/Step Functions, DynamoDB/RDS/Oracle, Docker/K8s/EKS/AKS, Json Schema, JOI, OpenAPI/Swagger,
Dec.,2020 – Dec., 2021 Yotaspale Cherry Hill, NJ (WFH),USA Staff Engineer	Lead Architect and Lead Implementation Engineer for Cloud(AWS/Azure/GCP) Cost Analytics/Optimization/Management platform OpenAPI/Swagger, Microservice, Java/Kotlin/SpringBoot, Kubernetes/EKS/AKS, Istio, Terraform, CQRS, Event Store, Event Driven Architecture (EDA), Kafka/ActiveMQ/IBM MQ, Kafka Stream, Docker, AWS(EC2/IAM/S3/ECS/EKS/ELB/Lambda/RDS/VPC/R53/Secret Manager/AutoScaling/SES...etc), Puppet/Chef/Ansible, Git/Maven/Jenkins, Python/Bash/Go, ELK/AppDynamic/Datadog,
Mar.,2020 – Nov., 2021 Gigster Inc. Cherry Hill, NJ,USA Technical Infrastructure/Devops Architect	Images manipulation/detection software evaluation system Microservice, NodeJs, Kubernetes/EKS, Terraform, RabbitMQ/ActiveMQ, Docker, AWS(EC2/IAM/S3/EKS/ALB/RDS/VPC/R53/AutoScaling/SES/ECR...etc), Puppet/Chef/Ansible, Git/Maven/Jenkins, JavaScript,
Jun.,2017 – Feb., 2020 Barclays Wilmington, DE,USA VP,Lead Architect Engineer	- Delivery Lead for strategic distributed system technology domain - Group-wide Cloud Adoption for Contact Center, Merchant Services and Fraud Terraform, CQRS, Event Store, Event Driven Architecture (EDA), Kafka/ActiveMQ/IBM MQ, Kafka Stream, Docker, AWS(EC2/IAM/S3/ECS/EKS/ELB/Lambda/RDS/VPC/R53/Secret Manager/AutoScaling/SES...etc), Puppet/Chef/Ansible, Git/Maven/Jenkins/Bamboo, Python/Ruby/Scala/Bash/Go, ELK/AppDynamic, DynamoDB/MongoDB/ElasticCache/GridGain, Linux/Windows/AIX/Solaris, aPaaS/VMWare/Citrix,
May, 2014 – June, 2017 Oracle Redwood, CA (WFH@NJ),USA Lead Software Engineer	- Docker Swarm/Compose/Stack CD/CI Test Environment for Cloud SDK/APIs/Connectors - Cloud Enterprise Applications Integration(EAI) connectivity SDK/API plugin framework for Oracle SOA Suite - Oracle Fusion Applications (FA) Business Event/Message-Driven Connector Framework Java, SOA, Web Services, SOAP/WSDL, REST/JSON/XML, XSchema/XSD, DOM, WSDL4J, SDK, API, MVC, Docker, Cloud, Amazon EC2, NoSQL, GIT, Maven, Junit, SoapUI, IaaS, SaaS, AWS,
Oct. 2010 – Apr. 2014 Comcast T&P Philadelphia, PA,USA Senior Cloud DevOps/Infrastructure Automation Engineer	- AutoBuild System for Build/Release more than 200 Microservice projects with GIT/SVN/Maven - Configuration management and deployment automation with Puppet and Capistrano - Continuous Integration (CI) and Continuous Deployment (CD) with Bamboo and Jenkins - Performance Monitoring System for Jetty, JVM, Linux and Splunk - Secured FTP server with virtual users and admins users with complex access permissions - BOM (Bill Of Materials) Administration and Management Application (BAMA) Enhancement and maintenance - Two Steps bare-metal Linux bootstrap system for HP Servers - Linux HA for Git Repo(Gerrit) and Maven Repo (Artifactory/Nexus) Servers - Automation scripts and tools for DevOps - CoreOS and Docker prototyping and testing - SDLC process management and junior team member training and mentoring DevOps, Automation, Docker, CoreOS, Ruby, Ruby on Rails 3, Bash, SDLC, GIT, Maven, Jira/Confluence, Bamboo/Jenkins/Hudson, Nexus/Artifactory, Gerrit, CI/CD, Performance Monitoring, Splunk, JMX, Capistrano, PXE/TFTP, DHCP, DNS, Cobbler, Centos, MySQL/PostgreSQL, NoSQL, Cassandra, RabbitMQ, JMS, LDAP, SSO, Metrics, Graphite, Agile, RPC, Cloud, Clustering, Linux HA, HAProxy, Pacemaker/Corosync/Cman, SAN, GFS2, RAC, Volume Manager,

Nov., 2006 – Oct., 2010 Boomi Inc. Berwyn, PA,USA Senior Software Engineer	• Boomi On-Demand Business Integration Cloud Service • Cloud Platform/Infrastructure Design and Implementation for Boomi On Demand Production Environment • Desktop Accounting and ERP Enterprise Applications Integration (EAI) • Amazon EC2 and S3 SOA HA Environment • Setup Development, Build and Deployment Environment for QA, Staging and Production Environment • Research virtualization technology and implement Virtual Machines using VMware server and workstation • Java to C/C++, DLL, COM/COM+/DCOM/.NET Integration • SaaS system integration • SAP system integration Java, SOA, IaaS, PaaS, EAI, XML, XSD, JAXB, JavaScript, .NET, EC2, S3, EBS, REST, SOAP, WSDL, SaaS, SAP, BAPI, ABAP, MySQL/PostgreSQL, Maven, SVN, JVM, JVM GC Tuning, Apache Lucene, Apache Solr, VMWare, Cloud,
Jun., 2006 – Nov.,2006 Vantage learning Newtown, PA,USA Senior System Engineer Manager	• Large scale SOA e-learning system architecture design and blueprinting • Large Data Center Design and Administration • Sun SPARCStorage SSA Array 100 and Sybase SQL Server (10.x, 11.x) Data (DB schema, data and Stored-Procedure) Recovery • Ecommerce Open Source Software installation and Configuration • Resin Application Server Distributed Session Design and Setup Linux, Centos, SOA, Java, .NET, Network, VLAN, Router, Firewall, Load Balancer, SOAP/WSDL, MySQL/PostgreSQL, Struts, Disk Array, Fedora, Redhat Linux, LAMP, eCommerce, SQL,
May, 2002 – Jun.,22,2006 Computer Science, Texas Tech University Lubbock, Texas,USA Senior Programmer/Analyst and Unix System Administrator	• Unix network System and Software routine maintenance • Setup Veritas Netbackup System with RAID0/1/5 and Tape Library • Setup LDAP, Kerberos KDC and Cross-Realm system • Migrated email system from UNIX mail server to Windows Exchange Server • Secured Postfix SMTP server with SASL Authentication and Verification • Installed AIX 5.1 and configured NIM on IBM RS6000 (10x 7043-140 boxes and 1x 7024-E30 box) • Deployed Globus Toolkit3.2.1 (OGSA/OGSI) Grid Computing Environment on Solaris 9(3 boxes) and Linux (3 boxes) • Setup Jumpstart Environment for Solaris 9 and Solaris 10 • Designed and Implemented the Single Sign-On (SSO) for Unix and Windows Environments • The Migration from NIS+ to LDAP for Solaris JAVA, C/C++, GCC, GNU Make, perl, LDAP, NFS, SSO, Kerberos, SSL, SMTP, Email Server and Client, Solaris, AIX, Mac OSX, Linux, Windows Server, PAM, RAID, SCSI, Veritas, TFTP, RARP/ARP, Bootparamd, DHCP, Postfix, PostgreSQL, NIS/NIS+,
Aug.,2002 – Jun.,2006 Computer Science, Texas Tech University Lubbock, Texas,USA Master Student	• Master Thesis: A Service Wrapping and Provisioning Framework for SOA • GIS Project: A Yield Mapping and Prediction Information Delivery System • Independently finished practical course projects list Java, C/C++, GCC, Gnu Make, Linux, Linux Kernel, GIS, Ant, Struts, J2EE, UML2, MVC, Maven, PostGIS, PostgreSQL, Solaris,

SELECTED PROJECTS

1. Feb.,2025 – Current, Director of Cloud Architecture, Biophy , Cherry Hill, NJ (WFH),USA

As a hands-on director of cloud architecture, I designed and implemented comprehensive infrastructure as code (IaC) by importing existing AWS resources and redesigning them according to AWS best practices. I led the development of a data ingestion and scraping system for an AI-powered RAG platform, following SOLID principles and Clean Architecture patterns. Additionally, I created a custom AWS CloudWatch custom alerting system leveraging Lambda, SQS and SNS for robust monitoring and notification.

1.1 Comprehensive Infrastructure as Code (IaC) Design and Implementation

Led the transformation of AWS infrastructure to infrastructure as code (IaC), importing and redesigning resources to align with best practices. Developed modular Terraform configurations and established secure, standardized cloud operations.

Technologies:

Terraform, AWS EC2/EBS/S3/VPC/ALB/Security/R53/IAM/Secrets Manager, Infrastructure as Code, AWS Best Practices, Security Controls,

1.2 AI-Powered Data Ingestion and Scraping System for RAG

Architected and delivered a robust data ingestion and scraping platform for AI-powered RAG, applying SOLID and Clean Architecture principles to ensure scalability, maintainability, and high data quality.

Technologies:

Python/Django/Pydantic, openAI API/LangChain, SOLID Principles, Clean Architecture, Data Processing, Web Scraping,

1.3 Custom AWS CloudWatch Alerting System with EventBridge, Lambda, SQS and SNS

Developed a proactive monitoring and alerting system using AWS CloudWatch, Lambda, SQS, and SNS, enabling automated detection and response to critical events across cloud infrastructure.

Technologies:

AWS CloudWatch, SQS/SNS, Lambda, EventBridge, Monitoring, Alerting,

2. Dec.,2021 – Feb.,2025, Senior Cloud Engineer II (Associate), McKinsey & Company , Cherry Hill, NJ (WFH),USA

Led the development of genAI systems with RAG and multi-agents to accelerate mainframe migration to modern technologies. Built comprehensive genAI solutions with RAG to speed up data pipeline development on PySpark and Databricks for complex data transformations. Scaled RAG vectorstore (ChromaDB) for a call center chatbot copilot. Successfully migrated and refactored complex EDA mainframe systems to AWS serverless architecture using EventBridge, Lambda, Step Functions, SQS, SNS, CloudWatch, and EKS.

2.1 Using genAI (openAI) with RAG (ChromaDB, Milvus) to accelerate the development of data pipelines running on pyspark and databricks

Designed and implemented a genAI system with RAG (ChromaDB/Milvus) to automate and accelerate the development of data pipelines, enabling seamless transformation of source tables to target tables with varying data structures.

Technologies:

Langchain/VectoreStore/ChromaDB/Milvus, genAI/openAI, RAG, Python/Pydantic/Asyncio/Flask/FastAPI/Debugpy, Typescript/JavaScript/NodeJs, Microservice, Dockerfile/Docker/Compose, PostgreSQL,

2.2 Using genAI (openAI) with multi-agents to accelerate the migration of mainframe to modern technologies

Built a multi-agent genAI platform to automate and accelerate the migration of mainframe applications to modern technologies, integrating RAG for context-aware transformation and process automation.

Technologies:

Langchain/VectoreStore/ChromaDB/Milvus, genAI/openAI, RAG, Agents/Multi-agents/Autogen, Python/FastAPI, Typescript/JavaScript/NodeJs, Microservice, Dockerfile/Docker Compose, PostgreSQL,

2.3 Migration/Refactoring of Mainframe (IBM MQ) EDA system to AWS by using EventBridge/Lambda/Step Function/CloudWatch/EKS/SQS/SNS...etc.

Engineered and deployed an EKS-based router to integrate AWS EventBridge with external MQ events, providing robust monitoring, reporting, and cloud-native migration for mainframe EDA systems.

Technologies:

Java/SpringBoot, Terraform, Microservice, Java/JavaScript, EKS, AWS/EventBridge/CloudWatch/Lambda, XML/XSD/Json/Json Schema, DynamoDB, RDS/Oracle,

3. Dec.,2020 – Dec., 2021, Staff Engineer, Yotascale , Cherry Hill, NJ (WFH),USA

Led the architecture and development of a comprehensive cloud cost analytics, optimization, and management platform supporting AWS, Azure, and GCP. Designed and implemented cross-cloud cost visibility, allocation, and optimization capabilities to help organizations manage cloud spending effectively.

3.1 Lead Architect and Lead Implementation Engineer for Cloud(AWS/Azure/GCP) Cost Analytics/Optimization/Management platform

Architected and implemented a cross-cloud cost analytics and optimization platform, enabling organizations to gain actionable insights and manage cloud spending across AWS, Azure, and GCP. Led the design of scalable data pipelines, analytics microservices, and secure infrastructure.

Technologies:

Cloud Cost Analytics/Optimization/Management, OpenAPI/Swagger, Microservice, Java/Kotlin/Python, Spring/SpringBoot/Flask, Kubernetes/EKS/AKS/GKS, Terraform/Terragrunt/Ansible, AWS/EC2/EBS/S3/VPC/Security/R53/IAM/Secrets Manager/Lambda, Service Mesh/Istio/Envoy, Helm/Chart, RabbitMQ, Azure/GCP, Junit/Test Containers, Docker/Minikube,

4. Mar.,2020 – Nov., 2021, Technical Infrastructure/Devops Architect, Gigster Inc. , Cherry Hill, NJ,USA

Designed and implemented comprehensive AWS infrastructure architecture including CI/CD automation, Terraform infrastructure as code, and EKS container orchestration. Led the development of scalable, automated cloud infrastructure solutions to support microservices and distributed systems.

4.1 Images manipulation/detection software evaluation system

Led the architecture and implementation of an image manipulation and detection evaluation platform using microservices and event-driven architecture on AWS EKS. Automated infrastructure provisioning and deployment pipelines for scalable, efficient operations.

Technologies:

Cloud Cost Analytics/Optimization/Management, OpenAPI/Swagger, Microservice, Java/Kotlin/Python, Spring/SpringBoot/Flask, Kubernetes/EKS/AKS/GKS, Terraform/Terragrunt/Ansible, AWS/EC2/EBS/S3/VPC/Security/R53/IAM/Secrets Manager/Lambda, Service Mesh/Istio/Envoy, Helm/Chart, RabbitMQ,

5. Jun.,2017 – Feb., 2020, VP,Lead Architect Engineer, Barclays , Wilmington, DE,USA

Directed the cloud adoption journey, migrating on-premises enterprise systems to AWS. Served as delivery lead for distributed system technology, including microservices, event-driven architecture, message transport, integration (batch/ETL), and event stream processing.

5.1 Delivery Lead for strategic distributed system technology domain

Directed the architecture and technology strategy for distributed systems, microservices, event-driven architecture, and stream processing, driving innovation and technology transformation across the organization.

Technologies:

Technology strategic Design and Stewardship, Distributed system, Domain Driven Design, Microservice, ACID/BASE, (Event) Stream /Batch Processing, Event Driven Architecture (EDA), Message Transports (Kafka, ActiveMQ, IBM MQ), Data Integration (ETL/ELT), Compensating transactions/Sagas, JMS/AMQP, Resilience and Performance, CQRS/SOA, CAP,

5.2 Group-wide Cloud Adoption for Contact Center, Merchant Services and Fraud

Led cloud assessment and migration strategy for hundreds of on-premises applications, providing technical guidance and troubleshooting to ensure successful adoption of AWS cloud architecture.

Technologies:

AWS, EC2, RDS, S3, ECS, VPC, ACL/Security Group/Routing table, ELB/NLB/ALB, Private Link, EBS,

6. May, 2014 – June, 2017, Lead Software Engineer, Oracle , Redwood, CA (WFH@NJ),USA

The Cloud Connectivity software engineering team is responsible to develop the Cloud Integration Connectivity SDK and Java Connectors which enable the SOA Suite and Oracle Integration Cloud Service (ICS) connectivity to any Oracle internal or external SaaS or On-Premises Enterprise Applications.

6.1 Docker Swarm/Compose/Stack CD/CI Test Environment for Cloud SDK/APIs/Connectors

Pioneered the use of Docker Swarm, Compose, and Stack to create scalable, automated CI/CD test environments for cloud SDKs, APIs, and Java connectors, streamlining development and testing workflows.

Technologies:

Docker, Docker Machine, Docker Compose, Docker Swarm, Docker Stack, Docker CLI (Rest APIs), WebLogic, Oracle Cloud VMs, AWS, JMS, NoSQL (Cassandra), DBMS (MySQL and PostgreSQL),

6.2 Cloud Enterprise Applications Integration(EAI) connectivity SDK/API plugin framework for Oracle SOA Suite

Developed a robust SDK and plugin framework for seamless integration between Oracle SOA Suite and a wide range of SaaS and on-premises enterprise applications, enhancing connectivity and extensibility.

Technologies:

Java, SOA, SOAP/REST, WSDL/JSON, XML, XML Schema, API, SDK, IaaS, RCU, JCA, XDK, JAXP, WSDL4J, DOM, WS-Security, Jersey REST/JSON, MVC, ADF, NLS/L10, Oracle DB, Jira, Agile, ADE, GIT, Maven, Junit, XMLUnit, SoapUI, OSC, ERP, HCM, RightNow,

6.3 Oracle Fusion Applications (FA) Business Event/Message-Driven Connector Framework

Engineered a business event and message-driven connector framework for Oracle Fusion Applications, enabling real-time event discovery, filtering, and integration with external systems.

Technologies:

Java, FA, OSC/ERP/HCM, WSDL/SOAP, REST, JSON, XSI, XML Schema, XDK, JAXP, WSDL4J, DOM, WS-Security, Jersey REST/JSON, MVC, ADF, ADE, Maven, Junit, XMLUnit, SoapUI,

7. Oct. 2010 – Apr. 2014, Senior Cloud DevOps/Infrastructure Automation Engineer, Comcast T&P , Philadelphia, PA,USA

As a senior engineer in BITT(Build Integration Test Team), provided DevOps support for multiple development teams and supporting the developers for SDLC infrastructure system design/implementation, and Build/Release/Deployment/SCM/CI automation., designed, developed and implemented the AutoBuild system, OpenSource performance monitoring system for the SDLC systems, two steps bare-metal Linux bootstrap system for HP servers, Linux HA Clusters for SCM/Maven/CI/Crowd servers, Ruby/Rails software system maintenance and improvement. Managed the SCM(Gerrit/Subversion), CI(Bamboo/Jenkins) and maven repo (Nexus and Artifactory) system. Designed and developed automation scripts (bash/ruby) to improve work efficiency. Led and Trained junior team members.

7.1 AutoBuild System for Build/Release more than 200 Microservice projects with GIT/SVN/Maven

Developed and automated the AutoBuild system to streamline build, release, and dependency management for over 200 microservice projects, reducing release cycles from days to hours.

Technologies:

Ruby/Rails, gems, Bash, Maven, GIT, SVN, Bamboo/Jira/Confluence CLI, Nexus/Artifactory REST/XML, SDLC, CI/CD, Gerrit/Subversion, MySQL/PostgreSQL, Crowd/OpenID, Capistrano,

7.2 Configuration management and deployment automation with Puppet and Capistrano

Implemented configuration management and deployment automation using Puppet and Capistrano, eliminating manual errors and accelerating application delivery across environments.

Technologies:

Puppet, Capistrano, Ruby, bash, CI/CD, GIT, Maven, Perl, Python,

7.3 Continuous Integration (CI) and Continuous Deployment (CD) with Bamboo and Jenkins

Standardized and automated CI/CD pipelines for 200+ microservices using Bamboo and Jenkins, improving release reliability and developer productivity.

Technologies:

CI/CD, Bamboo, Jenkins, Ruby, Bash, Python, GIT, Maven, Jira/Confluence,

7.4 Performance Monitoring System for Jetty, JVM, Linux and Splunk

Designed and integrated a comprehensive performance monitoring system for JVM, Linux, and Splunk, enabling proactive detection and resolution of system bottlenecks.

Technologies:

Metrics, graphite, jmx, splunk, Ruby, splunk-ruby-SDK, performance monitoring, UDP, Capistrano, god, bash, gdash, graphite, statsd, metricsd,

7.5 Secured FTP server with virtual users and admins users with complex access permissions

Engineered a secure FTP/SFTP environment with advanced user and group access controls, automating user management and improving security compliance.

Technologies:

ftp, sftp, chroot, bash, linux, user/group unix permission, sshd,

7.6 BOM (Bill Of Materials) Administration and Management Application (BAMA) Enhancement and maintenance

Enhanced and maintained the BAMA system for automated deployment and management of microservices, introducing customizable deployment features and integration with CI/CD tools.

Technologies:

Ruby, RoR 3, Rails 3, ActiveRecord, ActionView, ActionController, ActionMailer, Websocket, linux, JSON, Capistrano,

7.7 Two Steps bare-metal Linux bootstrap system for HP Servers

Developed a two-phase, fully automated Linux server bootstrap system for HP hardware, enabling zero-touch provisioning and configuration.

Technologies:

TFTP, PXE, DHCP, DNS, cobbler, RPC, xml, bash, ILO 3, IPMI, LVM, HP Servers, CentOS 6,

7.8 Linux HA for Git Repo(Gerrit) and Maven Repo (Artifactory/Nexus) Servers

The two nodes Linux HA cluster is for Git repo (Gerrit) / Maven Repo(Artifactory and Nexus) master/slave failover system. Using pacemaker/corosync/cman or keepalived/haproxy, it setup Gerrit, artifactory/Nexus master/slave automatically failover cluster with PostgreSQL/RAC and GFS2 over SAN / DRBD as backend DBMS and distributed file system.

Technologies:

Pacemaker, Corosync, Cman, Keepalived, HAProxy, GFS2, DRBD, SAN, Gerrit, Artifactory, Nexus, PostgreSQL, OpenID, LDAP, Crowd,

7.9 Automation scripts and tools for DevOps

Created a suite of automation tools and scripts to streamline DevOps workflows, improve efficiency, and reduce manual toil across the SDLC.

Technologies:

Ruby, Bash, Python, Capistrano, Jira/Rally, Bamboo/Confluence, Git/Gerrit, puppet, Nexus replication, REST/XML,

7.10 CoreOS and Docker prototyping and testing

Led the design and implementation of dynamic development and integration test environments using CoreOS and Docker, reducing setup time and increasing productivity.

Technologies:

CoreOS, Docker, Ruby, REST, LXC, cobbler, pxe, tftboot, DHCP,

7.11 SDLC process management and junior team member training and mentoring

Managed SDLC processes and mentored junior engineers, standardizing build, release, and deployment practices for over 200 projects.

Technologies:

SDLC, SCM, Git/Subversion, GIT Hooks, Ruby, Bash, Confluence, Jira, Agile,

8. Nov., 2006 – Oct., 2010, Senior Software Engineer, Boomi Inc., Berwyn, PA, USA

As the market and technology leader in on-demand integration, Boomi is the industry's first and leading integration platform-as-a-service and a two-time SIIA CODiE award winner - Best Application Integration Solution (2010) and Best On-Demand Platform (2009).

8.1 Boomi On-Demand Business Integration Cloud Service

Designed and developed a scalable, on-demand SOA integration platform enabling visual business process design, automated deployment, and seamless integration with desktop, ERP, and SaaS systems.

Technologies:

Java, XML, XSD, JAXB, REST, HTTP, Hibernate, openlaszlo, SSL, PKI, com4j, .NET 3.0/2.0/1.1, COM/DCOM, Linux, QXML, eConnect, servlet, Jetty, IDEA, Maven, subversion, Confluence, Jira, MySQL, RAID, SQL server, QuickBooks 2005, Sage MAS 90/200/500, MS Dynamics GP 9.0/10.0/2010, SOA,

8.2 Cloud Platform/Infrastructure Design and Implementation for Boomi On Demand Production Environment

Architected and managed a distributed, high-availability production environment for Boomi's cloud platform, including clustering, monitoring, and migration of critical infrastructure.

Technologies:

64-bits Redhat Enterprise Linux, MySQL clustering, Apache, PHP, Jetty, 64 bits Java, Apache Lucene/ Solr, Openlaszlo, distributed cache, distributed and parallel algorithm, ehcache, comet, continuation, DWR, JVM tuning, jmap, jstat, Sun JVM Heap, GC, vmstat,

8.3 Desktop Accounting and ERP Enterprise Applications Integration (EAI)

Engineered integration solutions for leading desktop accounting and ERP systems, enabling automated business data exchange and process automation.

Technologies:

QXML RequestProcessor, Microsoft eConnect 9.x/10.x/11, Librados, SAP R3, BAPI, REST/XML, COM/COM+/DCOM, .NET, MS Dynamics Navision, MS Dynamics CRM, SOAP/WSDL, WCF, XMLPort, C/AL, CodeUnits,

8.4 Amazon EC2 and S3 SOA HA Environment

Built and maintained a high-availability SOA environment on AWS EC2 and S3, implementing clustering, load balancing, and automated backup for robust cloud operations.

Technologies:

Fedora 6/7/8, AWS, EC2, S3, Jetty 6.x/7.x, MySQL 5.x, Apache httpd 2.x and Solr 1.3.x/1.4.x, NFS, shell script(bash and sh), ehCache, cron,

8.5 Setup Development, Build and Deployment Environment for QA, Staging and Production Environment

Established and automated development, build, and deployment environments for QA, staging, and production, streamlining release management and system administration.

Technologies:

Maven 2.x, subversion, TeamCity, PHP 5.x, Apache Http server 2.3, Apache Ant, MySQL5.x, Jetty 6.x, Openlaszlo 3.4, Apache Solr, Fedora 8,

8.6 Research virtualization technology and implement Virtual Machines using VMware server and workstation

Researched and implemented virtualization solutions using VMware, optimizing resource utilization and performance for development and testing environments.

Technologies:

Virtualization, paravirtualization, VMware Workstation 6.2 and VMware Server 1.4, Xen, Microsoft Virtual PC, Fedora 8, RAID, SMP, Performance tuning, Subversion, Maven, PHP, MySQL,

8.7 Java to C/C++, DLL, COM/COM+/DCOM/.NET Integration

Developed seamless integration between Java applications and native Windows technologies (C/C++, DLL, COM/DCOM, .NET), enabling cross-platform interoperability.

Technologies:

Java, .NET 3.0/2.0/1.1, COM/DCOM, Com4j, iKVM, C#, VB.NET, JNI, JNA, Win32 DLL, COMAdmin,

8.8 SaaS system integration

Designed and delivered integration solutions for leading SaaS platforms using REST/XML and SOAP/WSDL, expanding product connectivity and automation.

Technologies:

REST/XML, SOAP/WSDL, namespace, DOM, soapUI,

8.9 SAP system integration

Engineered robust integration between Java and SAP systems using web services and SAP JCO, enabling automated business process integration.

Technologies:

SAP R3, NetWeaver, Web Services, SOAP, WSDL, wsdl4j, SAP-JCO, RFC, BAPI, ABAP, Java,

9. Jun., 2006 – Nov., 2006, Senior System Engineer Manager, Vantage learning , Newtown, PA, USA

As the leader in cost-effective, high-volume, secure, scalable online assessment and instructional programs for K-12 and higher education, Vantage Learning leverages technology such as artificial intelligence, natural language understanding, and web-based learning objects. Vantage Learning has received accolades ranging from the prestigious CODIE Award for best instructional technology.

9.1 Large scale SOA e-learning system architecture design and blueprinting

Architected a scalable SOA-based e-learning platform leveraging Java/.NET and clustered PostgreSQL/MySQL databases, enabling secure, high-volume online assessment and instruction.

Technologies:

SOA, E-Learning, Struts, J2EE, .NET 2.0/3.0, Resin, SOAP, WSDL, UDDI, XML, PostgreSQL, MySQL,

9.2 Large Data Center Design and Administration

Designed and managed a mission-critical data center with 100+ servers and enterprise network devices, supporting 30+ applications and millions of user connections with high availability.

Technologies:

Solaris 5,7, Linux(Redhat,Fedora), F5 Big-IP load balancers, Switch, Router, VLAN, UPS, RAID 1,0,5, Windows Server, Terminal server, VNC, SPARCStorage, Tape Drive, Veritas, Apache, Resin, PostgreSQL,

9.3 Sun SPARCStorage SSA Array 100 and Sybase SQL Server (10.x, 11.x) Data (DB schema, data and Stored-Procedure) Recovery

Recovered critical patent data and database assets from failed Sun SPARCStorage SSA Array and Sybase SQL Server, ensuring business continuity and data integrity.

Technologies:

Solaris 5,7, Linux(Redhat, Fedora), Sybase SQL Server 10.x, Sybase ASE 11.x, 12.x 15.x, SSH, SPARCStorage SSA Array 100, Tape Drive,

9.4 Ecommerce Open Source Software installation and Configuration

Deployed and configured open source e-commerce platforms (osCommerce, x-cart, Sales-n-Stats) on PHP/MySQL, standardizing installation for rapid business enablement.

Technologies:

Linux(Redhat, Fedora), PHP5.x, MySQL 5.x, Apache 2.x,

9.5 Resin Application Server Distributed Session Design and Setup

Implemented distributed session management and HTTP load balancing on Resin Application Server, enhancing scalability and reliability for enterprise web applications.

Technologies:

Linux(Redhat, Fedora), resin 3.0, Java, Servlet, distributed session, cluster, load balancer (software and hardware),

10. May, 2002 – Jun.,22,2006, Senior Programmer/Analyst and Unix System Administrator, Computer Science, Texas Tech University., Lubbock, Texas, USA

System Administrator for networked Unix(Solaris, AIX, Mac OSX and Linux) and windows servers with SSO for Kerberos and LDAP, Veritas Netbackup for Disk Array with RAID 0/1/5, Migration from NIS/NIS+ to LDAP and Postfix/Sendmail configuration for the research and teaching lab for computer science department.

10.1 Unix network System and Software routine maintenance

Performed daily administration and maintenance of Solaris, AIX, and Mac OSX network systems, ensuring reliability and security across research and teaching labs.

Technologies:

Solaris 8,9,10, Linux, AIX, NFS, NIS/NIS+, LDAP, SMTP, IMAP, POP3, WebMail, SSH, Samba, Kerberos, Netgroup, Jumpstart, User Accounts, Quota, License server, GNU Make, GCC, Java, RAID, Tape Library, Veritas, Apache, MySQL, Sun Ray Server, PAM, Security, Syslog,

10.2 Setup Veritas Netbackup System with RAID0/1/5 and Tape Library

Implemented enterprise backup and recovery solutions for large RAID storage arrays and virtual file systems using Veritas NetBackup and tape libraries.

Technologies:

Veritas NetBackup Enterprise Server 5.1, Veritas NetBackup Client, RAID 0,1,5, SCSI, Volume Manager, VFS, Tape Library,

10.3 Setup LDAP, Kerberos KDC and Cross-Realm system

Designed and deployed a secure Single Sign-On (SSO) system with LDAP, Kerberos KDC, and cross-realm authentication, unifying access across Unix and Windows platforms.

Technologies:

SASL-2.1, GSSAPI, SEAM 1.2, SunOne Directory Server 5.2, NIS+, LDAP V3, PAM, Solaris 9, Windows Server 2003, Active Directory,

10.4 Migrated email system from UNIX mail server to Windows Exchange Server

Led the migration of departmental email from Unix servers to Windows Exchange, configuring secure SMTP/IMAP/POP3 with SSL for diverse clients.

Technologies:

Sendmail, Postfix, Unix Mail Client, Outlook, Solaris 9, Linux, Windows Server, Exchange Server, IMAP/S, POP3/S, SSL,

10.5 Secured Postfix SMTP server with SASL Authentication and Verification

Configured Postfix SMTP with SASL authentication and verification, enhancing email security for Unix and Linux environments.

Technologies:

Openssl-0.9.7, SASL-2.1, Postfix 2.4, SquirrelMail, pine, Solaris 9, Linux,

10.6 Installed AIX 5.1 and configured NIM on IBM RS6000 (10x 7043-140 boxes and 1x 7024-E30 box)

Installed and configured IBM AIX with NIM and NFS on RS6000 systems, enabling automated provisioning and dynamic resource management.

Technologies:

AIX 5.1, NIM, NFS, NIS/NIS+, SORCER/Jini/Rio,

10.7 Deployed Globus Toolkit 3.2.1 (OGSA/OGSI) Grid Computing Environment on Solaris 9(3 boxes) and Linux (3 boxes)

Deployed and secured Globus Toolkit 3.2.1 grid computing environments on Solaris and Linux, supporting distributed research workloads with SSL certificates.

Technologies:

Unix (Solaris 9, Redhat Linux 9), NFS, NIS/NIS+, Perl, PKI, X509, GNU Tools, Globus Toolkit 3.2.1, OGSA/OGSI, GSI,

10.8 Setup Jumpstart Environment for Solaris 9 and Solaris 10

Engineered automated Solaris deployments using Jumpstart, supporting multi-subnet installations and custom Perl scripts for NIS+ and tftp management.

Technologies:

Unix(Solaris 9), Jumpstart, TFTP, Bootparamd, NIS+, Perl, Security, snoop, RARP/ARP,

10.9 Designed and Implemented the Single Sign-On (SSO) for Unix and Windows Environments

Integrated Unix and Windows environments with a unified SSO solution using LDAP, Kerberos, AFS, and Samba, streamlining authentication and access control.

Technologies:

Unix(Debian Linux, Solaris 8 and 9), LDAP(OpenLDAP2.1.2, SunONE Directory Server 5.2 , MS Active Directory), AFS (OpenAFS 1.2.10), Kerberos (SEAM 1.2, MIT Kerberos V 1.3.1). PAM, GSSAPI, SSL, SASL, SSH, C++, GNU make, GCC, Visual.NET, Perl,

10.10 The Migration from NIS+ to LDAP for Solaris

Migrated Solaris NIS+ directory services to LDAP, designing custom schemas and automation scripts to modernize user management and authentication.

Technologies:

Unix(Solaris 8 and 9), LDAP(SunONE Directory Server 5.1), Kerberos (SEAM 1.2). NIS/NIS+, SSL, SSH, Perl, GNU make , GCC,

11. Aug.,2002 – Jun.,2006, Master Student, Computer Science, Texas Tech University , Lubbock, Texas,USA

Studied as Master Student in the CS department while working as part time as System Administration for the Teaching and Research labs for Unix/Linux and Windows environments

11.1 Master Thesis: A Service Wrapping and Provisioning Framework for SOA

Developed an automated SOA framework using Jini/Rio to wrap legacy applications as dynamic, provisioned services in a distributed grid environment, including a GIS-based yield tracker system.

Technologies:

Jini 2.0, RIO, JERI, Globus Toolkits 3.x, J2EE, Web Service, OpenGIS, PostGIS, JDBC, GeoServer, Struts, Ant, Tomcat, Servlet, JSP, UML2,

11.2 GIS Project: A Yield Mapping and Prediction Information Delivery System

Built a web-based GIS platform integrating OpenGIS, GeoServer, and PostGIS to deliver real-time yield prediction maps for farmers using satellite and field data.

Technologies:

OpenGIS, PostGIS, PostgreSQL, JDBC, GeoServer, GeoTools, Struts, Ant, Apache Maven, Servlet, JSP, MVC,

11.3 Independently finished practical course projects list

Completed advanced projects including a C-based LL(1) compiler, UML-modeled supermarket system, parallel matrix multiplication service, Linux kernel modules, Java card applications, and mobile map navigation tools.

Technologies:

GCC Compiler, Solaris, Linux Kernel 2.4, UML 2, Rational Rose, Java, SORCER, SOA, Jini, JavaSpace, JBoss, J2EE, VFS, Java Card platform, MIDP, MIDlets, Servlets, OCF, CLDC, Ant,

HOBBIES PROJECTS:

H.1 Personalized JSON Schema Resume

Customized the open source project dynamic-json-resume using JSON format and mustache templates with Node.js and CSS3. GitHub url: <https://github.com/rongyj/dynamic-json-resume/tree/rongyj>

Technologies:

JavaScript, HTML5, CSS3, Node.js, JSON, mustache, Puppeteer,